

Curriculum Vitae

Current Position

Aug. 2022 – Present | **NeuroAI Scholar**, Cold Spring Harbor Laboratory, Long Island, NY
Independent postdoctoral scholar
Bridging computer science and neuroscience in artificial neural networks to build more energy-efficient computing. *Primarily interested in how evolution constrains intelligence.*
Also collaborating with neuroscientists to study facial expressions in mice.

Education

2016 – 2022 | **Ph.D. in Electrical Engineering**, University of Wisconsin-Madison,
Dissertation: "Building Energy Efficient Computers with Brain-Inspired Computing Models"

2016 – 2019 | **M.S. in Electrical Engineering**, University of Wisconsin-Madison,

2012 – 2016 | **B.S. in Computer Engineering and Cont. Applied Mathematics**, Rose-Hulman Institute of Technology,
Magna Cum Laude

Academic Experience

Research Experience

2016 – 2022 | **Research Assistant**, University of Wisconsin-Madison
Advised by: Dr. Mikko Lipasti
Worked on energy-efficient computing paradigms in stochastic computing and neural networks.

2014 – 2016 | **Research Assistant**, Rose-Hulman Institute of Technology
Advised by: Dr. Mario Simoni and Dr. Daniel Chang
Developed configurable architecture for simulating Hodgkin-Huxley neural systems.

Teaching Experience

Sep. 2021 – Dec. 2021 | **Lecturer**, University of Wisconsin-Madison
ECE252: Introduction to Computer Engineering
Instructed a course section on introductory material for computer engineering.

Sep. 2018 – Aug. 2019 | **Teaching Assistant**, University of Wisconsin-Madison
ECE532: Matrix Methods for ML
Taught as an in-class TA for a flipped class on linear algebra and machine learning.

Sep. 2017 – Dec. 2018 | **Teaching Assistant**, University of Wisconsin-Madison
ECE315: Intro. to Microprocessor Laboratory
Instructed lab course on designing, assembling, and programming a printed circuit board.

Jan. 2017 – May 2018 | **Teaching Assistant**, University of Wisconsin-Madison
ECE353: Intro. to Microprocessor Systems
Taught as an in-class TA for a flipped class on embedded systems.

Mar. 2016 – May 2016 | **Teaching Assistant**, Rose-Hulman Institute of Technology
ECE530: Advanced Microcomputers

Mar. 2014 – May 2014 | **Teaching Assistant**, Rose-Hulman Institute of Technology
CSSE332: Operating Systems

Manuscripts

Preprints (in-progress work)

- May 2025 **Walking the Weight Manifold: a Topological Approach to Conditioning Inspired by Neuromodulation**, *Preprint*
Authors: A. S. Benjamin*, K. Daruwalla*, C. Pehle*, A. Zador
- Nov. 2021 **Accelerating Deep Learning with Dynamic Data Pruning**, *Preprint*
Authors: R. S. Raju, K. Daruwalla, M. Lipasti

Publications

- Dec. 2026 **A Control-Theoretic Approximation to Predictive Coding Dynamics**, *NeurIPS 2026*
Authors: R. Fayyazi*, K. Daruwalla*, M. Javadzadeh
- Dec. 2026 **BEAST3D: Animal behavioral analysis and neural encoding from multi-view video via Gaussian splatting**, *NeurIPS 2026* (🗣️ oral)
Authors: Y. Wang, L. Aharon, W. Zhu, K. Daruwalla, L. Zhang, J. Zou, S. Chettih, H. Hou, L. Paninski, M. R. Whiteway
- Apr. 2026 **Cheese3D: Sensitive Detection and Analysis of Whole-Face Movement in Mice**, *Nature Neuroscience*
Authors: K. Daruwalla*, I. N. Martin*, L. Zhang, D. Naglič, A. Frankel, C. Rasgaitis, R. Zhao, X. Zhang, Z. Ahmad, J. C. Borniger, X. H. Hou
- Dec. 2024 **Continual learning with the neural tangent ensemble**, *NeurIPS 2024* (🔦 spotlight paper)
Authors: A. S. Benjamin, C. Pehle, K. Daruwalla
- Dec. 2024 **Delays in generalization match delayed changes in representational geometry**, *NeurIPS 2024 Workshop on Unifying Representations in Neural Models (UniReps)*
Authors: X. Zheng, K. Daruwalla, A. S. Benjamin, D. Klindt
- May 2024 **Information Bottleneck-Based Hebbian Learning Rule Naturally Ties Working Memory and Synaptic Updates**, *Frontiers in Computational Neuroscience*
Authors: K. Daruwalla, M. Lipasti
- Apr. 2024 **BitFit: Bitstream-Aware Training for Stochastic Neural Networks**, *Second Workshop on Unary Computing (WUC 24)*
Authors: N. Joshi, K. Daruwalla, M. Lipasti
- Feb. 2023 **Energy-Efficient Bayesian Inference Using Bitstream Computing**, *IEEE Computer Architecture Letters*
Authors: S. Khoram, K. Daruwalla, M. Lipasti
- Nov. 2019 **BitSAD v2: Compiler Optimization and Analysis for Bitstream Computing**, *ACM Transactions on Architecture and Code Optimization (TACO)*
Authors: K. Daruwalla, H. Zhuo, R. Shukla, M. Lipasti
- Jun. 2019 **BitBench: A Benchmark for Bitstream Computing**, *Conference on Languages, Compilers, and Tools for Embedded Systems (LCTES 19)*
Authors: K. Daruwalla, H. Zhuo, C. Schulz, M. Lipasti
- Jun. 2019 **Resource Efficient Navigation Using Bitstream Computing**, *First ISCA Workshop on Unary Computing (WUC 19)*
Authors: K. Daruwalla, H. Zhuo, M. Lipasti
- Jun. 2019 **BitSAD: A Domain-Specific Language for Bitstream Computing**, *First ISCA Workshop on Unary Computing (WUC 19)*
Authors: K. Daruwalla, H. Zhuo, M. Lipasti

Jan. 2019 **A quantitative analysis of the performance of computing architectures used in neural simulations**, *Journal of Neuroscience Methods*
 Authors: K. Daruwalla, N. Olivero, A. Pluger, S. Rao, D. W. Chang, M. Simoni

Presentations

Invited and contributed talks

Sep. 2025	Exploiting structure in brains and machines <i>CSHL Annual Postdoc Retreat</i> , Long Island, NY	[Invited]
Jun. 2023	Intro. to FluxML and Machine Learning in Julia <i>Data Umbrella Seminar Series</i> , Online (YouTube Live)	[Invited]
Feb. 2022	Building Energy-Efficient Computers <i>Cold Spring Harbor Lab NeuroAI Seminar</i> , Long Island, NY	[Invited]
Jan. 2020	BitSAD v2: A Domain-Specific Language for Bitstream Computing <i>High-performance Embedded Architecture and Compilation Conference (HiPEAC 20)</i> , Bologna, Italy	[Conference]
Oct. 2019	BitSAD v2 <i>Industry Affiliates Meeting</i> , Madison, WI	
Jun. 2019	Resource Efficient Navigation Using Bitstream Computing <i>First ISCA Workshop on Unary Computing (WUC 19)</i> , Phoenix, AZ	[Workshop]
Jun. 2019	BitBench: A Benchmark for Bitstream Computing <i>Conference on Languages, Compilers, and Tools for Embedded Systems (LCTES 19)</i> , Phoenix, AZ	[Conference]
Oct. 2018	BitSAD: A Domain-specific language for Bitstream Computing <i>Industry Affiliates Meeting</i> , Madison, WI	
Oct. 2017	Seeing Through the FoG: A Biologically Inspired Navigation System <i>Industry Affiliates Meeting</i> , Madison, WI	

Posters

May 2026	Extending Cheese3D with Soft Body Curves to Measure Facial Movements in Mice <i>AI in Biology Symposium</i> , Long Island, NY	[Conference]
Jul. 2025	Neuromodulation implies a manifold of model weights <i>NeuroAI in Seattle 2025</i> , Seattle, Washington	[Conference]
Mar. 2025	Cheese3D: sensitive detection and analysis of whole-face movement in mice <i>Computational and Systems Neuroscience (COSYNE 2025)</i> , Montreal, QC	[Conference]
Dec. 2024	Continual learning with the neural tangent ensemble <i>NeurIPS 2024</i> , Vancouver, Canada	[Conference]
Sep. 2024	Generative modeling of trained networks as an analogy for neuronal development <i>From Neuroscience to Artificial Intelligence (NAISys 2024)</i> , Long Island, NY	[Conference]
Mar. 2024	The dynamics of interpretable 3D facial features reflect hidden neural and physiological states in mice <i>Neuronal Circuits</i> , Long Island, NY	[Conference]
Nov. 2021	A Biologically Plausible Learning Rule Based on the Information Bottleneck <i>Spiking Neural networks as Universal Function Approximators (SNUFA 21)</i> , Virtual	[Workshop]

Oct. 2019	BitSAD v2 <i>Industry Affiliates Meeting, Madison, WI</i>	
Jun. 2019	BitBench: A Benchmark for Bitstream Computing <i>Conference on Languages, Compilers, and Tools for Embedded Systems (LCTES 19), Phoenix, AZ</i>	[Conference]
Oct. 2018	BitSAD: A Domain-specific language for Bitstream Computing <i>Industry Affiliates Meeting, Madison, WI</i>	
Oct. 2017	Seeing Through the FoG: A Biologically Inspired Navigation System <i>Industry Affiliates Meeting, Madison, WI</i>	
Nov. 2016	Drone Control with Map-Seeking Circuits <i>Industry Affiliates Meeting, Madison, WI</i>	

General public talks

Jun. 2024	Cocktails & Chromosomes: Kyle Daruwalla <i>Cocktails & Chromosomes Series, Long Island, NY</i>	
Apr. 2024	Research & AI Panel Discussion <i>Long Island Artificial Intelligence Conference, Long Island, NY</i>	[Panel]
Jun. 2023	Cocktails & Chromosomes: Kyle Daruwalla <i>Cocktails & Chromosomes Series, Long Island, NY</i>	
Jan. 2023	Computers, Brains, and the In-Between <i>Cold Spring Harbor Lab DeMystifying Science Series, Long Island, NY</i>	

Awards

Sep. 2021	Second place – AFRL xView2 Overhead Imagery ML Hackathon
May 2018	Gerald Holdridge Teaching Excellence Award
Feb. 2016	Honor Student Award

Service, Mentorship, and Leadership

Mentorship

Research mentor	• Catherine Rasgaitis (undergraduate) • Irene Nozal Martin (graduate student)	• Xingyu Zhang (graduate student)
Open-source mentor	• Abhirath Anand • Jiangeng (as part of Google Summer of Code for FluxML)	• Chandra Kiran • Saksham Rastogi

Service and Leadership

Program Committee	Second Workshop on Unary Computing, Third Workshop on Unary Computing
Reviewer	NeurIPS, ICLR, UniReps, Workshop on Unary Computing
2021 – 2022	Graduate student representative on departmental committee
2019 – 2020	Vice-President of ECE Graduate Student Association
2017 – 2019	President of ECE Graduate Student Association
2016 – 2017	Public Relations Officer of ECE Graduate Student Association
2012 – 2016	President of Linux Users Group

2014 – 2016 | Operations Manager of Rose Tech Radio Club
2015 – 2016 | Founder of Rose Maker Lab

Press

Jun. 2026 | **CSHL's Kyle Daruwalla and Irene Nozal Martin create Cheese3D for mouse faces**, *Times of Huntington-Northport*, 7 June 2026. Available: <https://tbrnewsmedia.com/cshls-kyle-daruwalla-and-irene-nozal-martin-create-cheese3d-for-mouse-faces/>
Jun. 2024 | **Can AI learn like us?**, *CSHL Press*, 20 June 2024. Available: <https://www.cshl.edu/can-ai-learn-like-us/>
Jul. 2024 | **Advancing AI's boundaries at Cold Spring Harbor Labs**, *Long Island Herald*, 5 July 2024. Available: <https://www.liherald.com/stories/advancing-ais-boundaries-at-cold-spring-harbor-labs,208824>

Industry Experience

Jun. 2020 – Dec. 2020 | **Research Co-op Intern**, AMD, Austin, TX
Explored neural network sensitivity to input perturbations in the context of scientific computing.
May 2017 – Aug. 2017 | **Digital Design Intern**, Texas Instruments, Dallas, TX
Performed IP design for multimedia IPs.
Jun. 2016 – Aug. 2016 | **Digital Design Intern**, Texas Instruments, Dallas, TX
Performed design verification for high speed I/O IPs.
Sep. 2015 – May 2016 | **Project Intern**, Rose-Hulman Ventures, Terre Haute, IN
Worked with project team to design various products for several clients.
Jun. 2015 – Aug. 2015 | **Digital Design Intern**, Texas Instruments, Dallas, TX
Performed design verification for high speed I/O IPs.
Sep. 2013 – May 2015 | **Project Intern**, Rose-Hulman Ventures, Terre Haute, IN
Worked with project team to design various products for several clients.